

Homework 1

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1. Team Members

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2. Dataset Description

The Framingham Heart Study is a long running longitudinal study, beginning in 1948, that has followed multiple generations over time for the purpose of determining risk factors of cardiovascular diseases. This dataset is a subset of patients between 1956 to 1968. Patients were then followed for 24 years to determine if they had any of the following events: Angina Pectoris, Myocardial Infarction, Atherothrombotic Infarction or Cerebral Hemorrhage (Stroke) or death. We have a total sample size of $n = 3206$ after filtering for subjects who have 3 or more visits (originally had $n = 4434$). Follow-up visits occurred every 6 years, so our measurements occur at baseline, 6 years, and 12 years. <https://search.r-project.org/CRAN/refmans/riskCommunicator/html/framingham.html>

3. Variables

Continuous Outcome

The continuous outcome we will be studying is systolic blood pressure (mmHG). At each visit, blood pressure was taken three times. The value stored in this variable is the average systolic blood pressure of the last 2 measurements.

Binary Outcome

The binary outcome we will be studying is coronary heart disease (CHD). This is defined as any of the following conditions: Angina Pectoris, Myocardial infarction (Hospitalized and silent or unrecognized), Coronary Insufficiency (Unstable Angina), or Fatal CHD.

Grouping Variable

The grouping variable we will use is sex. There are 2 groups: male and female.

Other Variables

- Age at exam
- BMI
- Smoking Status (current/not current smoker)
- Total cholesterol (mg/dL)
- Diabetes
- Use of Anti-hypertensive medication at exam
- Heart rate (beat/min)
- Glucose (mg/dL)

4. Descriptive Summary

See next page.

5. Limitations and Challenges

One challenge we may face has to do with the spacing between data points. While patients came every 6 years, it was not exactly constant for all patients. We have a variable that measures the number of days from baseline exam to follow up. For the 1st follow up the range of days is between 1734 and 2562 days (4.75 to 7.02 years). For the 2nd follow up the range of days is between 3899 and 4854 days (10.68 to 13.29 years). This may potentially cause us challenges.

Another challenge we may face has to do with our binary outcome variable. These patients, while all having 3 follow-up visits within the dataset, were followed for 24 years to determine if they developed any of the adverse outcomes being study. Because of this, we don't know when our outcome occurred, just that it eventually did occur. A patient has a 1 for all 3 visits if they developed the outcome at any point within the study or during the 24 years of follow-up. Alternatively, there is also a binary indicator that tells us if a patient has the condition during a specific visit. We could also use this variable for our outcome depending on how the methods we use require the outcome to be set up.

Lastly, the distribution of systolic blood pressure is notably right-skewed across all periods and for both sexes, though the skew seems to be higher for females than it is for males, on average. We will have to take this skew into account and carefully handle outliers when conducting our analyses.

```
framingham2 = framingham2 %>% mutate(  
  PERIOD = PERIOD %>% as.factor(),  
  SEX = if_else(SEX==1,"Male","Female"))
```

```
ggplot(framingham2, aes(x = PERIOD, y = SYSBP, fill = PERIOD)) +  
  geom_boxplot() + facet_grid(~SEX) +  
  ggtitle("Distributions of Systolic Blood Pressure by Period") +  
  xlab("Period") + ylab ("Systolic Blood Pressure (mmHg)") +  
  theme(plot.title = element_text(hjust = 0.5), legend.position = "none")
```

Table 1: Patient Baseline Characteristics

	Overall (N=3206)
Sex	
Female	1843 (57.5%)
Male	1363 (42.5%)
Missing	0 (0%)
Age	
Mean (SD)	48.7 (8.32)
Median [Min, Max]	48.0 [32.0, 70.0]
Missing	0 (0%)
Body Mass Index	
Mean (SD)	25.7 (3.98)
Median [Min, Max]	25.3 [15.5, 56.8]
Missing	7 (0.2%)
Smoking Status	
Nonsmoker	1681 (52.4%)
Smoker	1525 (47.6%)
Missing	0 (0%)
Total Cholesterol (mg/dL)	
Mean (SD)	235 (44.0)
Median [Min, Max]	232 [113, 696]
Missing	32 (1.0%)
Diabetes	
No	3148 (98.2%)
Yes	58 (1.8%)
Missing	0 (0%)
Use of Anti-Hypertensive Medication	
No	3086 (96.3%)
Yes	80 (2.5%)
Missing	40 (1.2%)
Heart Rate (beat/min)	
Mean (SD)	75.2 (11.8)
Median [Min, Max]	75.0 [44.0, 125]
Missing	0 (0%)
Casual Serum Glucose (mg/dL)	
Mean (SD)	80.9 (19.9)
Median [Min, Max]	78.0 [40.0, 386]
Missing	299 (9.3%)
Systolic Blood Pressure	
Mean (SD)	130 (20.1)
Median [Min, Max]	127 [83.5, 243]
Missing	0 (0%)
Incidence of Any CHD During Followup	
No	2378 (74.2%)
Yes	828 (25.8%)
Missing	0 (0%)

Distributions of Systolic Blood Pressure by Period

